PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: JadualIQ

Version: 1.0

1. Project Overview

- **Problem Statement:** Part-time workers in Malaysia juggle shift offers from multiple employers with no tool to help them decide which one is actually worth taking. Existing scheduling apps are built for employers, not workers; they ignore transport costs, travel time, and how one shift affects the rest of the week. The result is workers who accept shifts that lose them money on transport, commute time and scheduling conflicts. Decisions get made on gut feel, and the financial hit is invisible until payday
- **Target Domain:** AI-powered Decision Intelligence for multi-employer shift workers in Malaysia — part-time workers, students, and gig-flex staff who work shifts across multiple outlets or employers.
- **Proposed Solution Summary:** JadualIQ is a web-based multi agent AI tool where workers submit their available shift and personal constraints. Where 6 specialized tool agents gather real world signals in parallel – weather, traffic, transit, pay benchmark, fatigue and earnings gap. Z.AI's GLM then synthesises these into a ranked shift recommendation with plain-language explanations of every trade-off, allowing workers to maximise their net weekly earnings while respecting rest and budget constraints.

2. Background & Business Objective

- **Background of the Problem:** In Malaysia's retail and F&B sector, part-time workers often juggle life across multiple employers – like a student alternating between a mamak, a cinema, and a retail outlet. The core issue is a massive data disconnect where existing scheduling software is employer-centric. Employer A has no idea what Employer B is scheduling, the workers is the person who can see their entire week but they are forced to make split second scheduling decisions completely in their head. Because they cannot

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instantly calculate dynamic, real world variables like travel, sudden transport surges, or cumulative fatigue they end up guessing.

- **Importance of Solving This Issue:** The reliance on "gut-feel" scheduling creates income leakage where a worker might accept a shift because the hourly rate looks good only to lose all their profit to a rainy-day Grab surge or unpaid travel time. Currently shift workers lose an estimated 10% to 20% of their potential net earnings to these uncalculated blind spots. JadualIQ acts as the mathematical filter to stop the bad decisions from happening. By optimizing shifts for actual net yield instead of gross pay, the system can reliably recover between RM40 and RM120 per week for the average casual worker.

- **Strategic Fit / Impact:** JadualIQ targets the "AI for Economic Empowerment" by translating data into financial decisions. In this architecture, the Z.AI GLM is the mandatory synthesis engine. Without the GLM, our individual data agents(weather,traffic,fatigue) would just output a mess of disconnected numbers. The GLM is required to actually resolve complex trade offs such as high paying shifts against fatigue warning and output a single economically optimized decision that the user can immediately trust

3. Product Purpose

3.1. Main Goal of the System: [e.g. To provide an AI-enabled partner in design.]

3.2. Intended Users (Target Audience):

- Product Manager
- Founder of Products (Early Stage)
- Entrepreneurs

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- UI/UX Designer
- Frontend Developers
- Low Code Builders

4. System Functionalities

4.1. Description: [e.g.The system operates as a high-speed generative engine for design purposes through natural language. Using Prompt-to-UI model, intelligence captures required relationships, user flows, multi-screen layout and many more designs accordingly.]

4.2. Key Functionalities:

- **Vibe Designing: [e.g.**Translate human-language descriptive instructions into high-fidelity designs]
- **Conversational Editing:** [e.g.Facilitates continuous modification of the existing design by speaking to canvas itself/ texting.]
- **Versioning & Infinite Canvas:** [e.g. AI Agent Enabled version tracking with dozens of design variation generations is provided.]
- **Export- Component Based:** [e.g. Enables conversion of the visual into code snippets also editable Figma tool layers.]

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Do state the LLM used (Z AI GLM) along with justification why and what capabilities made it the right choice for the product's core functionalities.

4.3.2. **Prompting Strategy:** Describe your team's approach [e.g. zero-shot, few shot or multi-step agentic prompting]. So, explain the reason why the strategy chosen does fit your use case.

4.3.3. **Context & Input Handling**: Describe how the system that your team developed handles the unstructured or inputs that are oversized. So, what's the maximum input size your

product does accept before it actually does truncation or chunking or rejection? So, what's going to happen to those inputs that exceed the limit?

4.3.4. **Fallback & Failure Behavior:** What does the developed product do when the model returns off-topic or unusable and most importantly hallucinated response? So, is there any mechanism for retrying, a human escalation path or even a graceful error state?

5. User Stories & Use Cases

- **User Stories:** *Example:* "As a product manager, I want to enter a prompt using my own way of describing so that it provides me with a design without manual drawing by me."
- **Use Cases (Main Interactions):**

  Prompt to UI Flow: [e.g. Users enter the prompt, then Stitch parses the intent, identifies required components, and then renders a customizable 2D Canvas.]

  Flow Generation: [e.g. User selects a component, and triggers/enables the section called "Generate Next Screen". So, Stitch generates the target screen while maintaining the digital token i.e. color/fonts etc.]

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- [e.g. A conversational canvas for users to generate and edit in 2D workspace.]
- [e.g. Coherent and consistent design flow for entire system]
- [e.g. One click export feature along with codes.]
- [e.g. Support both text to UI & picture to UI feature]

7. Features Not Included (Scope Control)

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- [e.g. Implementation of backend logic, API, & database]
- [e.g. Only for frontend scaffolding, does not support production environment]
- [e.g. Multi-user inputs simultaneously are not supported [Initial Versions]]

8. Assumptions & Constraints

- **LLM Cost Constraint:** State your team's estimation on token cost per average user session. Do identify at least one design related decision that is made specifically to keep the inference costs manageable at scale. [e.g. caching repeated queries].
- **Technical Constraints:** [e.g. Code export features are only limited to export in HTML, CSS. Other modern frameworks like React/ Swift (IOS) are not supported.]
- **Performance Constraints:** [e.g. To design high-quality visuals, it requires many tokens, so there is a generation limit.]
- **User Input:** [e.g. The model is unable to generate without human prompts and is not ready for production so requires human approval.]

9. Risks & Questions Throughout Development

- **Design Similarity: [e.g.** How are we ensuring that AI is not generating similar designs across users?]
- **Frame Stability: [e.g.** Does the user need to fix the layout if it is found broken after exporting?]
- **Drawing Interpretation**: [e.g. What will happen if the AI is unable to contextualize the sketch that is provided?]

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